

Knowledge Distillation with Ternary Quantization-Aware Training: ResNet34 Teacher to Ternary ResNet18 Student on CIFAR-10

Tanmayee KM
ATDL Assignment I

1 Introduction

Modern convolutional networks such as ResNet achieve strong image classification accuracy but carry substantial memory and compute cost, largely dominated by the storage and arithmetic of their convolutional and fully-connected weights [1]. Reducing this cost without a large accuracy penalty is essential for deployment on memory- and power-constrained devices. Two complementary strategies address this: **weight quantization**, which reduces the numerical precision used to represent weights, and **knowledge distillation**, which transfers the predictive behavior of a large, accurate model into a smaller one.

This work combines both. We quantize a ResNet18 student’s convolutional and fully-connected weights to the ternary set $\{-\alpha, 0, +\alpha\}$ following the Ternary Weight Networks (TWN) formulation of Li & Liu (2016), as restated in Zhu et al. (2017) [3, 4], applying this constraint throughout training via Quantization-Aware Training (QAT) rather than as a post-hoc step. Because the ternary threshold operation is non-differentiable, gradients are propagated using a clipped Straight-Through Estimator (STE), whose foundational idea comes from Bengio, Léonard, and Courville (2013) [5] and whose convergence behavior for quantized networks is analyzed by Yin et al. (2019) [6]; the specific saturated/clipped variant implemented here follows the convention introduced for weight binarization by Hubara et al. (2016) [7]. Simultaneously, the ternary student is trained with Knowledge Distillation [2] from a frozen, full-precision ResNet34 teacher, combining temperature-scaled soft-target matching with standard hard-label supervision.

The objective of this study is to evaluate whether a heavily compressed, ternary-weight ResNet18 can retain competitive CIFAR-10 classification accuracy relative to both its full-precision ResNet18 counterpart and its ResNet34 teacher, and to quantify the accuracy/compression trade-off, the effect of the KD temperature hyperparameter, and the practical limitations of this approach.

Experimental setup, in brief:

ResNet34 (FP32) teacher $\rightarrow$ ResNet18 (FP32) baseline $\rightarrow$ ResNet18 (ternary) KD+QAT student.

The KD temperature T is the ablated hyperparameter, compared at $T = 4$ (main configuration) and $T = 2$, with the KD mixing weight fixed at $\lambda = 0.9$.

2 Methodology

2.1 Dataset and Experimental Setup

All models are trained and evaluated on CIFAR-10. The 50,000-image training set is split into 45,000 training and 5,000 validation images (fixed seed); the 10,000-image test set is held out entirely until final evaluation and is not used for model selection at any stage.

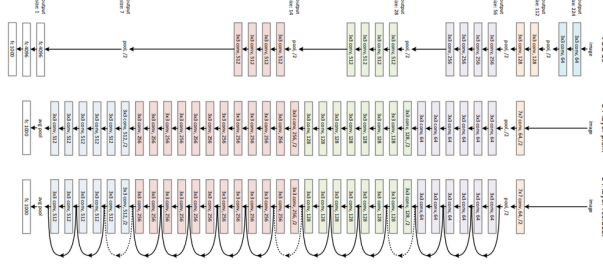


Figure 1: Architecture and methodology of the proposed approach: ResNet34 FP32 teacher, ResNet18 FP32 baseline, and ResNet18 ternary student trained using knowledge distillation and quantization-aware training.

Table 1: Hyperparameters

Setting	Value
Dataset	CIFAR-10 (45,000 train / 5,000 val / 10,000 test)
Random seed	134
Augmentation (train only)	Pad-4 random crop $\rightarrow 32 \times 32$, random horizontal flip, per-channel normalize
Batch size	128
Optimizer	SGD, momentum 0.9, weight decay $1e-4$
Initial learning rate	0.1
LR schedule	Step decay $\div 10$ at epochs 100 and 150
Total epochs	200 (all models)
Ternary threshold	$\Delta = 0.7 \times \text{mean}(W)$, per layer
Ternary scale	$\alpha = \text{mean } W_i $ over surviving weights
STE clip value	1.0
KD loss (main)	$T = 4, \lambda = 0.9$
KD loss (ablation)	$T = 2, \lambda = 0.9$

2.2 Model Architecture

All three models use a CIFAR-adapted ResNet stem (3×3 convolution, no initial max-pool), built from the residual BasicBlock of He et al. (2016) [1].

- **ResNet34 Teacher:** [3,4,6,3] BasicBlocks per stage, FP32, 21,282,122 parameters.
- **ResNet18 Baseline:** [2,2,2,2] BasicBlocks, FP32, no distillation, 11,173,962 parameters.
- **ResNet18 Ternary Student:** identical [2,2,2,2] layout; all convolutional and fully-connected weights ternary-constrained; BatchNorm parameters and all biases remain FP32.

2.3 Ternary Weight Quantization

Following the TWN formulation of Li & Liu (2016), as formulated and restated in Zhu et al. (2017, Eqs. 4–5) [3, 4], each layer’s full-precision weight tensor $\tilde{W}$ is converted to a ternary forward weight $W_q \in \{-\alpha, 0, +\alpha\}$ on every forward pass:

$$W_q = \begin{cases} +\alpha & \tilde{W} > \Delta \\ 0 & |\tilde{W}| \leq \Delta \\ -\alpha & \tilde{W} < -\Delta \end{cases}, \quad \Delta = 0.7 \times \text{mean}(|\tilde{W}|) \quad (1)$$

with α set to the mean absolute value of the weights that survive thresholding. Both Δ and α are computed **per layer** (pooled over the entire weight tensor, not per output channel), matching TWN’s own formulation. All convolutional and fully-connected layers are ternarized, including the first convolution and final fully-connected layer; BatchNorm scale/shift parameters and all biases remain FP32. This is the TWN formulation as implemented in this work; Zhu et al. (2017) additionally propose a learned-scale variant (TTQ), not used here.

The latent FP32 weight $\tilde{W}$ is retained and updated by the optimizer; only the forward pass uses W_q . Because the thresholding operation has zero gradient almost everywhere, backpropagation uses a **clipped/saturated Straight-Through Estimator**: the gradient with respect to $\tilde{W}$ equals the gradient with respect to W_q where $|\tilde{W}| \leq 1$, and is zeroed otherwise. This estimator follows the saturated-STE convention introduced for weight binarization by Hubara et al. (2016) [7]; its general rationale — that a bounded surrogate is more stable than an unconditional identity pass-through — is supported by analogy to the convergence analysis of Yin et al. (2019) [6], whose proof applies to activation quantization rather than weight quantization directly.

2.4 Knowledge Distillation + QAT

The student is trained with the temperature-scaled KD loss of Hinton, Vinyals, and Dean (2015) [2]:

$$L = (1 - \lambda) L_{CE} + \lambda T^2 L_{KL} \quad (2)$$

where L_{CE} is standard cross-entropy against the true label, and L_{KL} is the KL divergence between the teacher’s and student’s temperature-softened output distributions (p_t^T, p_s^T). The teacher (best checkpoint, ResNet34) is loaded in `.eval()` mode with all parameters frozen (`requires_grad=False`); its forward pass runs under `torch.no_grad()`, and zero gradient leakage into teacher parameters was explicitly verified. The student’s forward pass uses ternary weights throughout training (QAT, not post-hoc quantization); its latent FP32 weights are updated via the clipped STE described in Section 2.3. Main configuration: $T = 4$, $\lambda = 0.9$.

3 Training Results

3.1 Teacher Training

Training accuracy approaches 100% while validation accuracy plateaus around 94–95%, indicating a moderate generalization gap. The learning-rate reduction at epoch 100 produces a visible improvement in validation accuracy and a sharp drop in training loss, consistent with the model settling into a sharper minimum once the step size is reduced.

3.2 FP32 ResNet18 Baseline

This establishes the full-precision reference point against which the ternary student is evaluated, isolating the effect of ternarization+KD from any confound introduced by the smaller architecture alone.

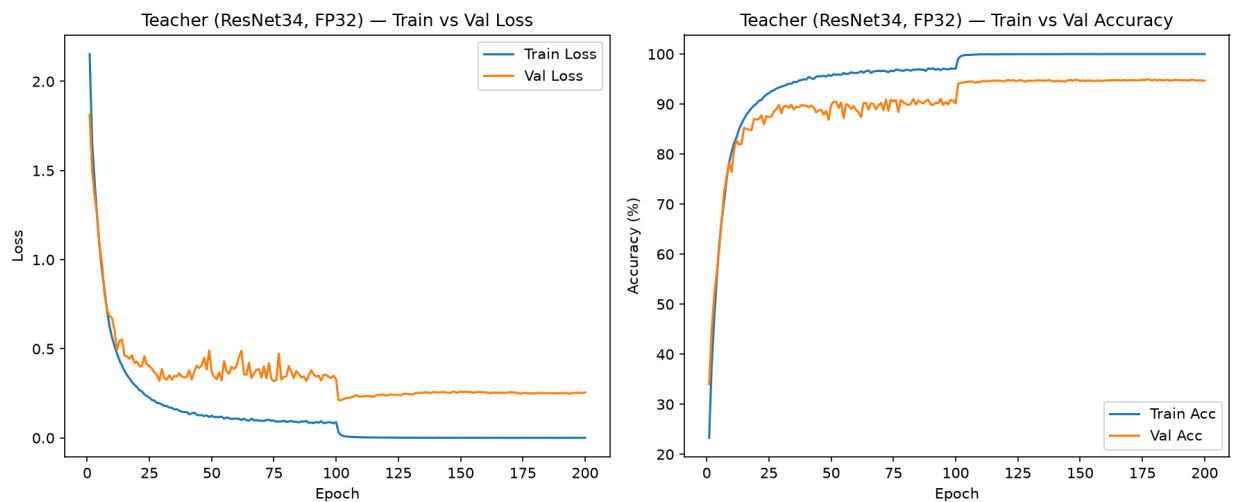


Figure 2: Teacher (ResNet34, FP32) — Train vs Validation Loss / Accuracy.

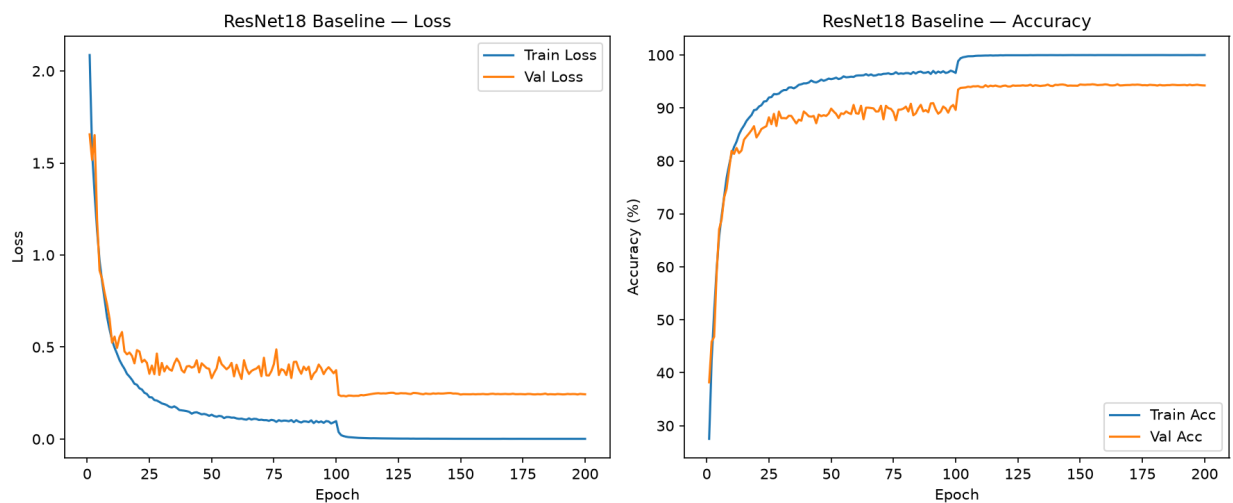


Figure 3: Baseline (ResNet18, FP32, no KD) — Train vs Validation Loss / Accuracy.

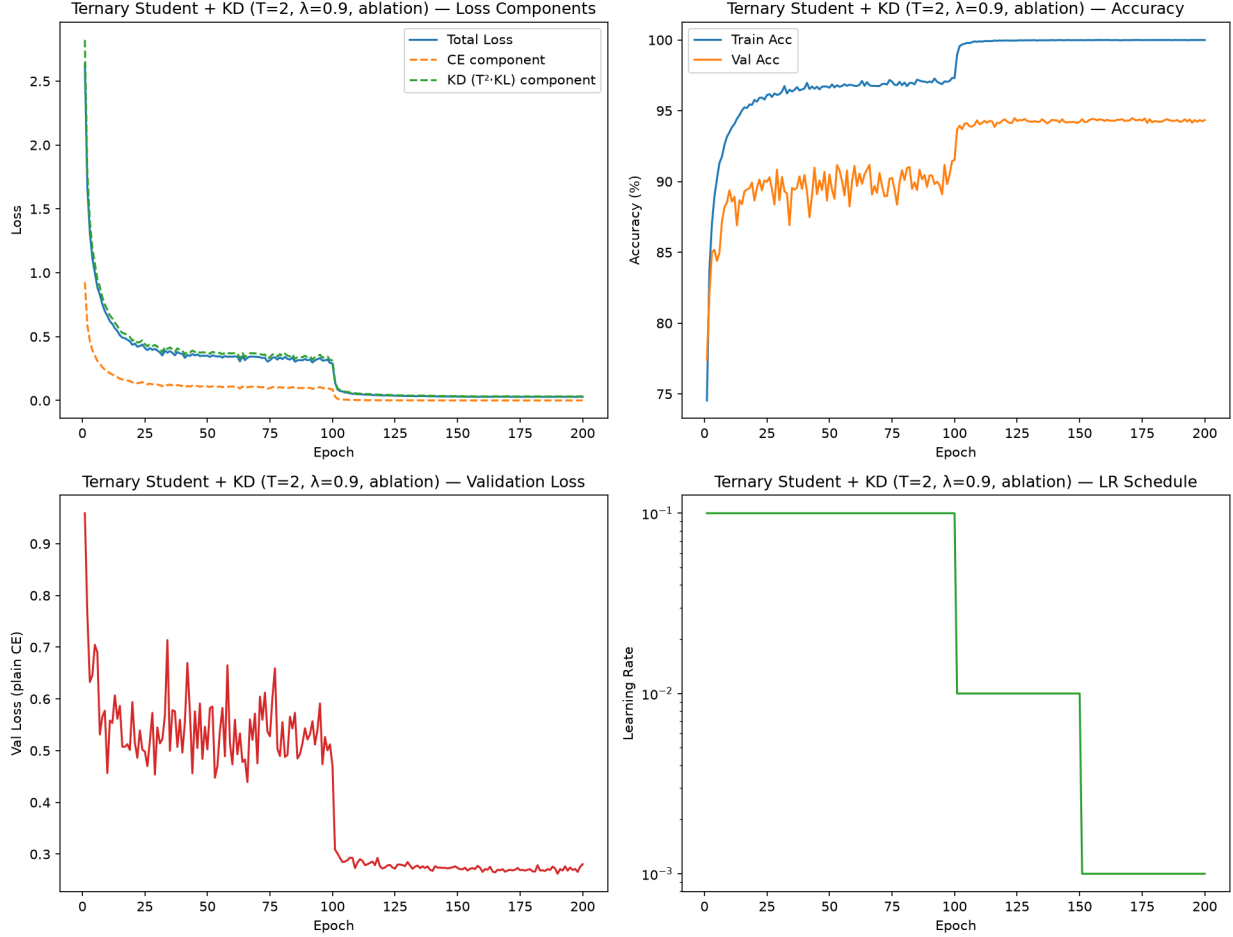


Figure 4: Ternary Student + KD ($T=2$, $\lambda=0.9$) — Loss Components / Accuracy / Validation Loss / LR Schedule.

3.3 Ternary Student Training

The student shows rapid initial learning and stable convergence under the ternary constraint, with training accuracy approaching 100% and validation accuracy settling around 94% after the epoch-100 LR reduction. This demonstrates that QAT combined with KD trains stably under the ternary constraint rather than diverging or stalling — a key rubric requirement (joint KD+QAT pipeline stability).

During the early epochs (before the first LR drop), the KD-trained ternary students reach higher validation accuracy *faster* than the FP32 baseline trained without KD (see Figure 4) — consistent with KD providing a denser, more informative training signal early in optimization, even though all models converge to similar final accuracy after epoch 100.

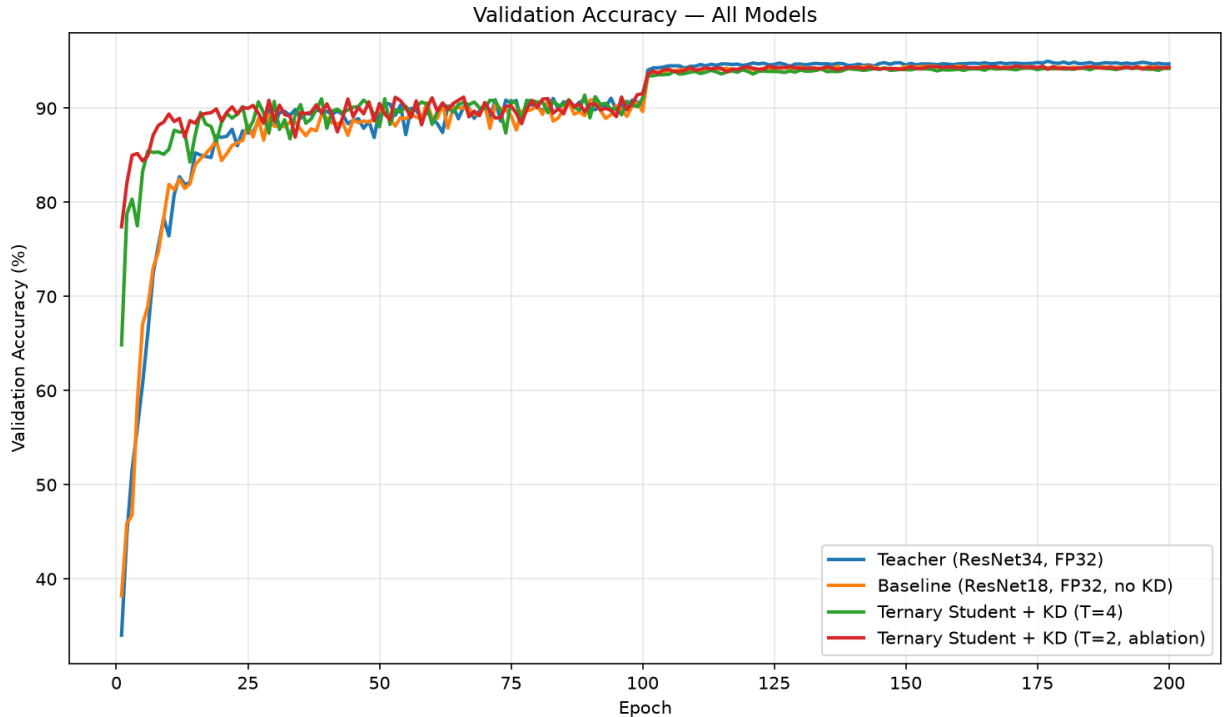


Figure 5: Validation Accuracy — All Models

4 Evaluation and Compression Analysis

4.1 Final Test Performance

Table 2: Main Evaluation

Model	Precision	Test Accuracy	Test Loss
ResNet34 Teacher	FP32	94.39%	0.2726
ResNet18 Baseline	FP32	94.13%	0.2582
ResNet18 Ternary + KD (T=4)	Ternary	94.29%	0.3455
ResNet18 Ternary + KD (T=2, ablation)	Ternary	94.28%	0.2935

The ternary student (94.29%) essentially matches both the FP32 baseline (94.13%) and sits within 0.10pp of the much larger FP32 teacher (94.39%), despite an $\sim 11\times$ reduction in trained parameters relative to the teacher and full ternary weight constraint. Test loss is higher for the ternary student than either FP32 model, which is expected: cross-entropy loss is more sensitive to confidence calibration than raw accuracy, and the ternary constraint plus KD’s soft-target training objective produce a differently calibrated output distribution even where the final class decision is correct.

4.2 Validation Accuracy Comparison

All four models converge to accuracy levels close to one another. The improvement after epoch 100 coincides with the learning-rate reduction, suggesting the lower LR enabled further refinement

across all models. Despite the severe reduction in weight precision, the ternary students do not exhibit substantial degradation in validation accuracy relative to the FP32 models.

4.3 Confusion Matrix and Per-Class Performance

Across all four models, confusion matrices show broadly similar error patterns, indicating that ternarization does not fundamentally alter the *type* of classification errors the network makes. The most persistent confusion is between cat and dog — the two visually and semantically closest classes in CIFAR-10 — across every model, including the full-precision teacher. Visually distinctive classes (automobile, ship, frog) achieve consistently higher accuracy across all four models. This indicates the ternary student retains a similar class-level decision structure to its FP32 counterparts, rather than developing qualitatively different failure modes.

4.4 Ternary Weight Distribution and Sparsity

Pooled across all 21 ternary layers (11,164,352 total ternary weights): $-\alpha$: 24.0%, 0: 59.2%, $+\alpha$: 16.9%.

$$\text{Global sparsity} \approx 59.2\% \quad (3)$$

The split between $-\alpha$ (24.0%) and $+\alpha$ (16.9%) is not symmetric, indicating the network’s latent weights are on average slightly negatively skewed across layers, even though the ternary scale magnitude α itself is symmetric by construction.

Layer-wise sparsity ranges roughly from 49% to 80% (simple unweighted layer average: 59.37%, close to but distinct from the true weighted global figure of 59.2%). The scale α varies substantially by depth — largest in the stem (first layer, $\alpha \approx 0.637$) and the final FC layer ($\alpha \approx 0.378$), smallest in the deepest residual stage ($\alpha \approx 0.026$ – 0.15). This indicates a single global scaling factor would not adequately represent the weight magnitude of every layer, and confirms the value of per-layer (rather than a single network-wide) α/Δ . This first-layer/last-layer-vs-middle-layer asymmetry echoes the layer-dependent behavior reported by Zhu et al. (2017) [4] for TTQ, though our implementation ternarizes the first and last layers rather than exempting them as TTQ’s own experiments do.

Table 3: Compression Summary

Metric	FP32 ResNet18	Ternary ResNet18
Total parameters	11,173,962	11,173,962
Ternary weights	—	11,164,352
FP32 parameters	11,173,962	9,610
Global sparsity	—	$\approx 59.2\%$
Theoretical storage	42.625 MB	2.699 MB
Compression	$1\times$	$\approx 15.8\times$ (vs. FP32 ResNet18); $\approx 30.08\times$ (vs. FP32 ResNet34 teacher, 81.185 MB)

Important distinction: the 2.699 MB figure is the *theoretical* storage if only the 2-bit ternary indices and per-layer FP32 α scales were serialized. The actual saved `.pth` checkpoint stores the student’s **latent FP32 weights** (needed to resume or re-verify training) and is therefore close in size to the FP32 baseline’s checkpoint — not 2.699 MB. This repository reports the theoretical deployment-time figure only; no separate compact export format is implemented.

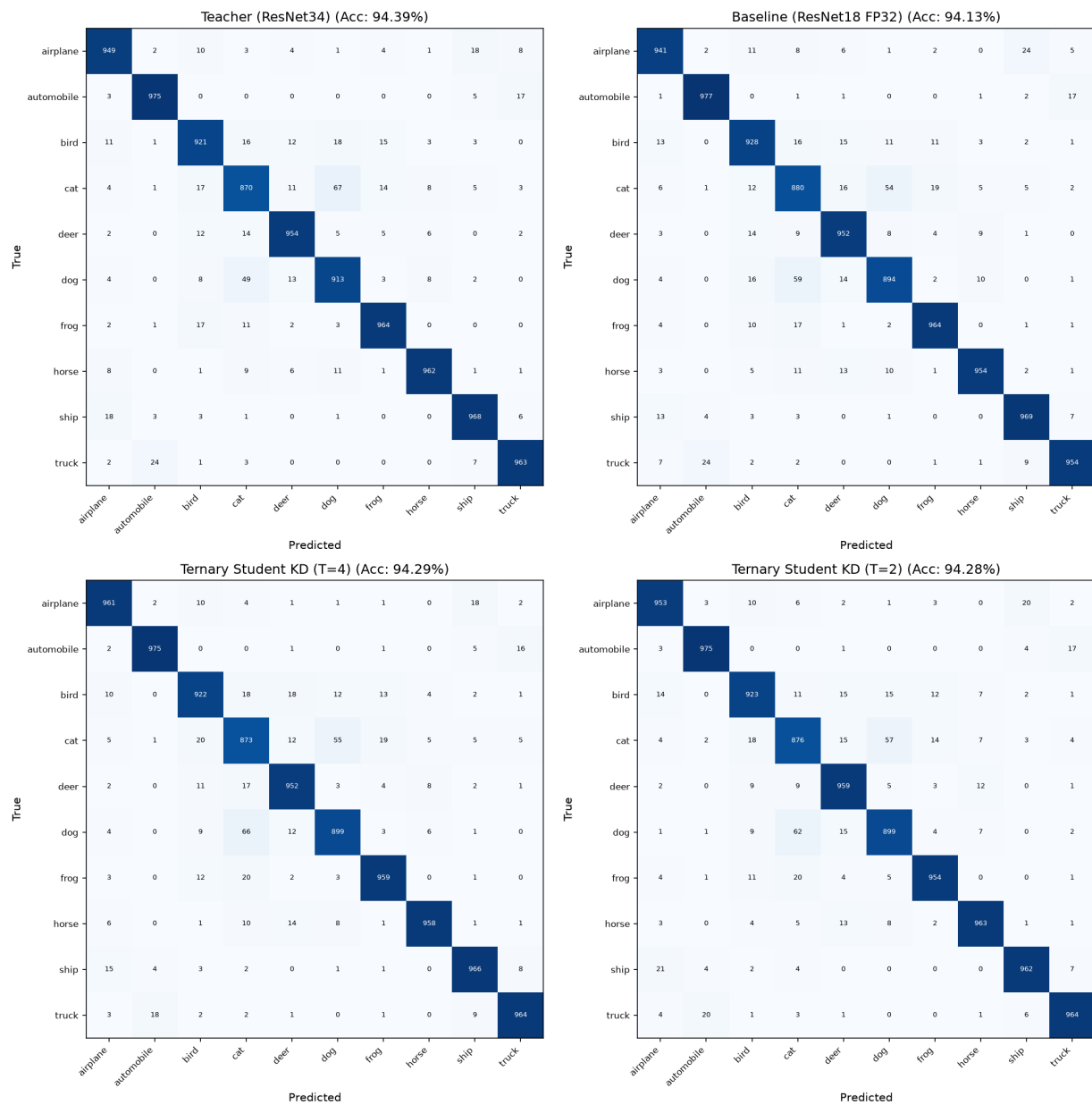


Figure 6: Confusion matrices — all four models.

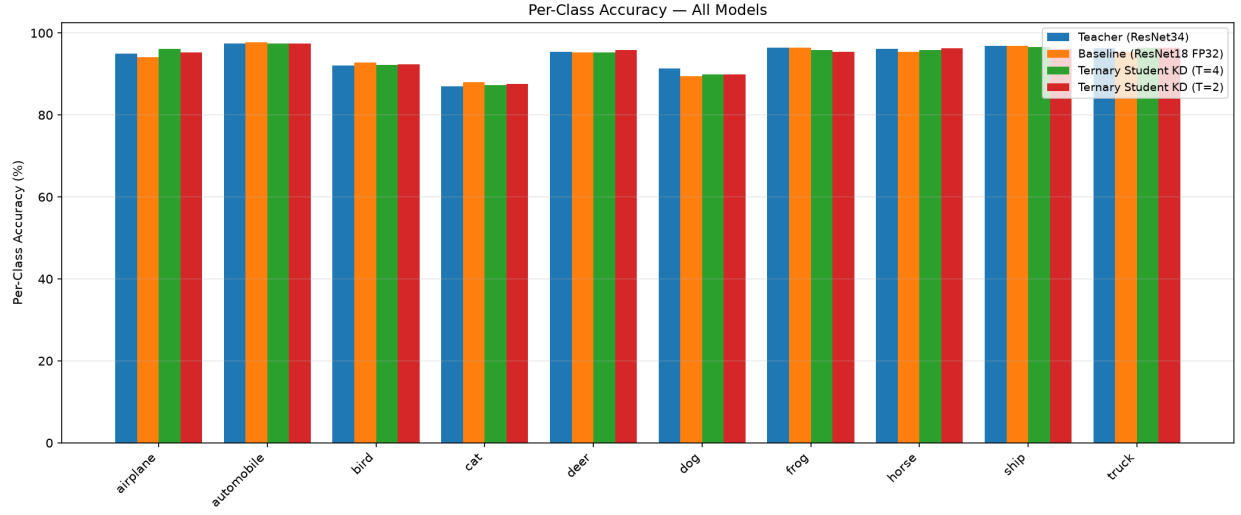


Figure 7: Per-class accuracy comparison — all four models

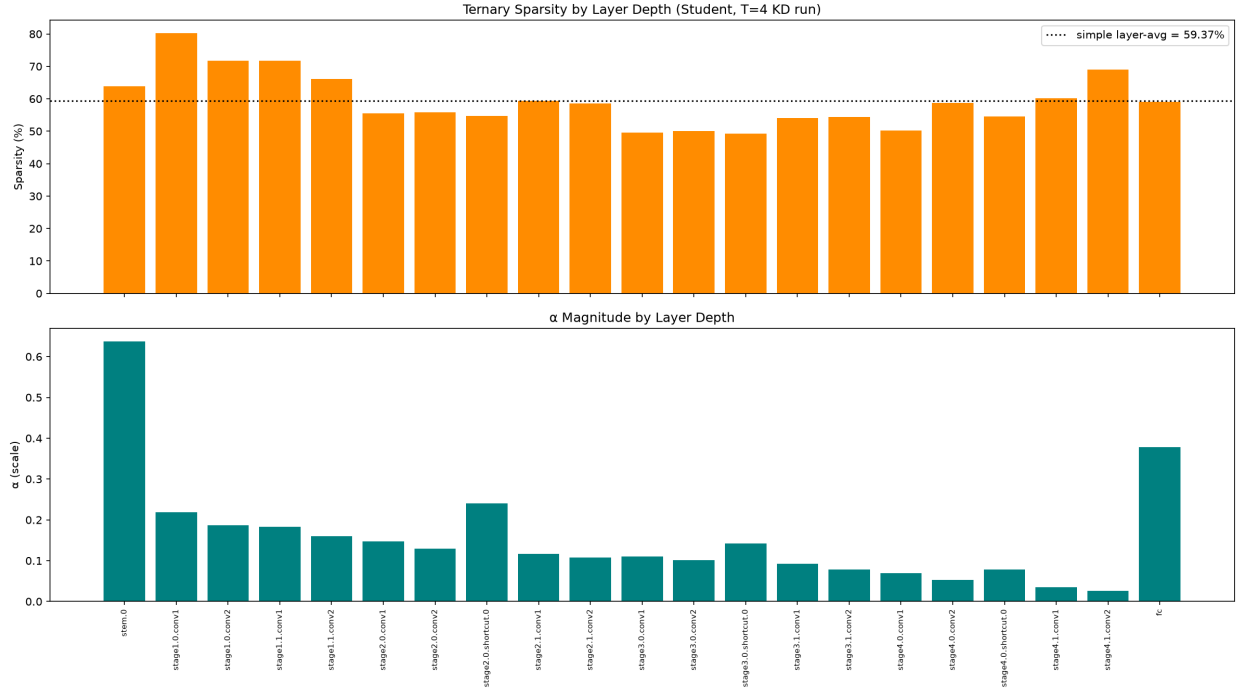


Figure 8: Ternary forward weight distribution, pooled across all 21 ternary layers.

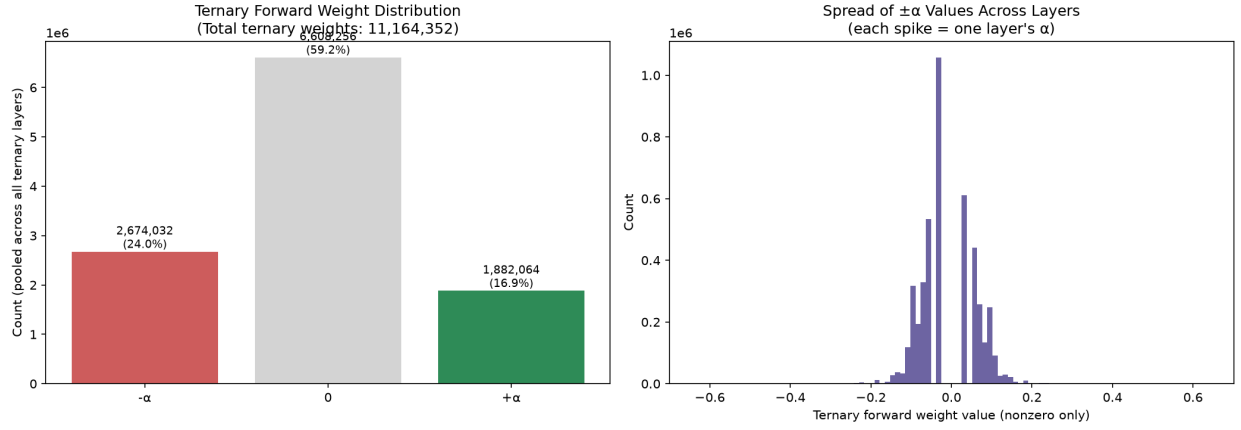


Figure 9: ternary forward weight distribution.png

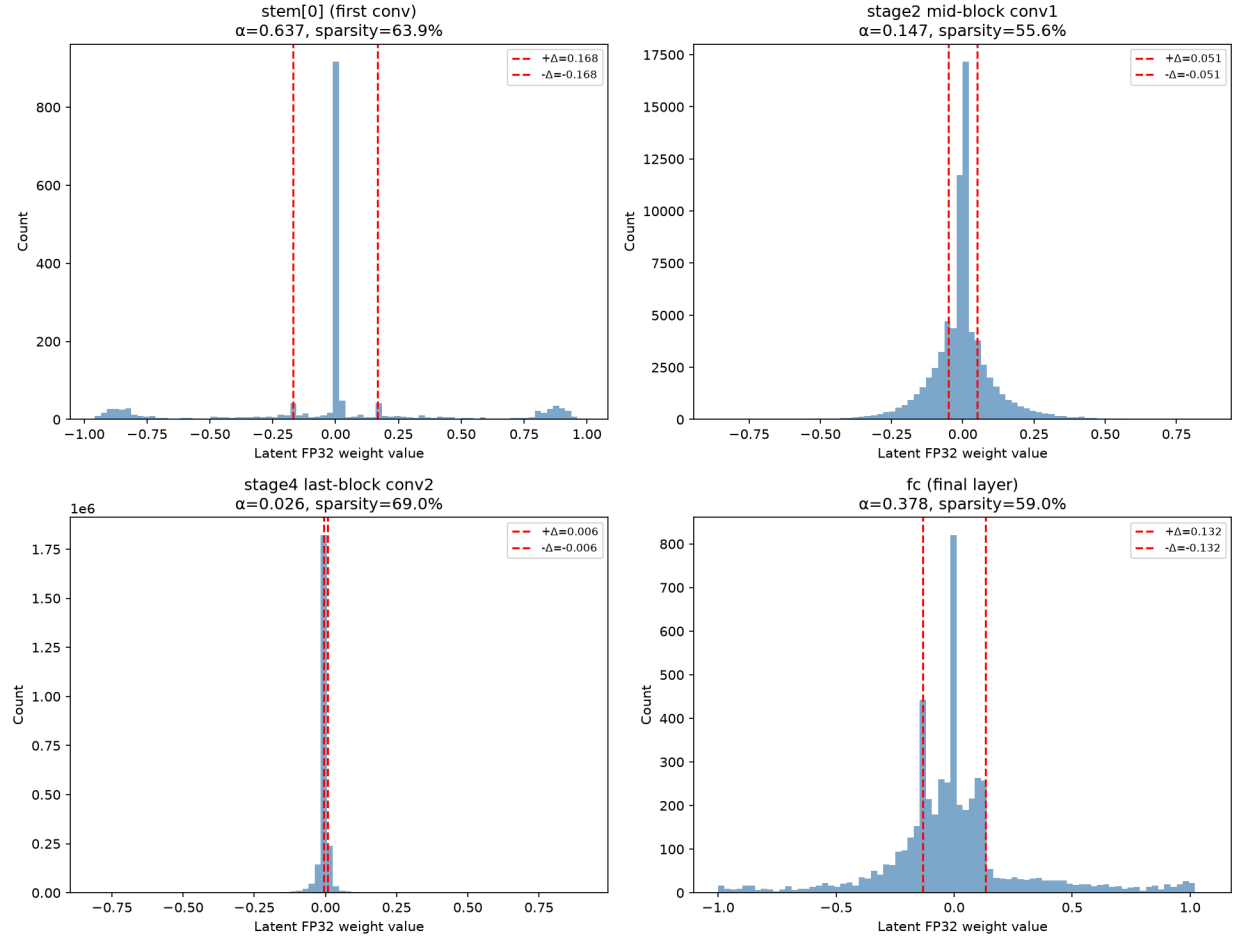


Figure 10: latent weight histograms

4.5 External Out-of-Distribution Evaluation (CIFARNet)

To assess generalization beyond the CIFAR-10 distribution, the main student checkpoint ($T=4$, $\lambda=0.9$) was evaluated on 100 images (10 per class) from the EleutherAI/CIFARNet test split — an ImageNet-derived dataset sharing CIFAR-10’s 10 class labels but sourced from different imagery, native 64×64 resolution (resized to 32×32 for this model). Only the main checkpoint was tested; the $T=2$ ablation checkpoint was not evaluated on this dataset.

Table 4: CIFARNet Out-of-Distribution Accuracy

Class	Accuracy
airplane	80.0%
automobile	80.0%
bird	90.0%
cat	60.0%
deer	80.0%
dog	50.0%
frog	90.0%
horse	70.0%
ship	90.0%
truck	90.0%
Overall	78.0%

Accuracy drops from 94.29% (in-distribution CIFAR-10 test) to 78.0% on this out-of-distribution set — an expected consequence of distributional shift (different image source, upsampled resolution) rather than a flaw in the ternary/KD approach specifically. The weakest classes, dog (50%) and cat (60%), are also the pair with the highest mutual confusion in the in-distribution confusion matrix (Section 4.3), suggesting the model’s already-least-confident decision boundary degrades most under distribution shift. Only 10 images per class were tested, so these per-class figures carry high sampling variance and should be read as indicative rather than statistically precise.

5 KD Temperature Ablation

5.1 Experimental Setup

Only the KD temperature T is varied ($T = 4$ main vs. $T = 2$ ablation); $\lambda = 0.9$ and every other hyperparameter (Table 1) is held identical.

5.2 Results

Table 5: Ablation Results

T	λ	Test Accuracy
4	0.9	94.29%
2	0.9	94.28%

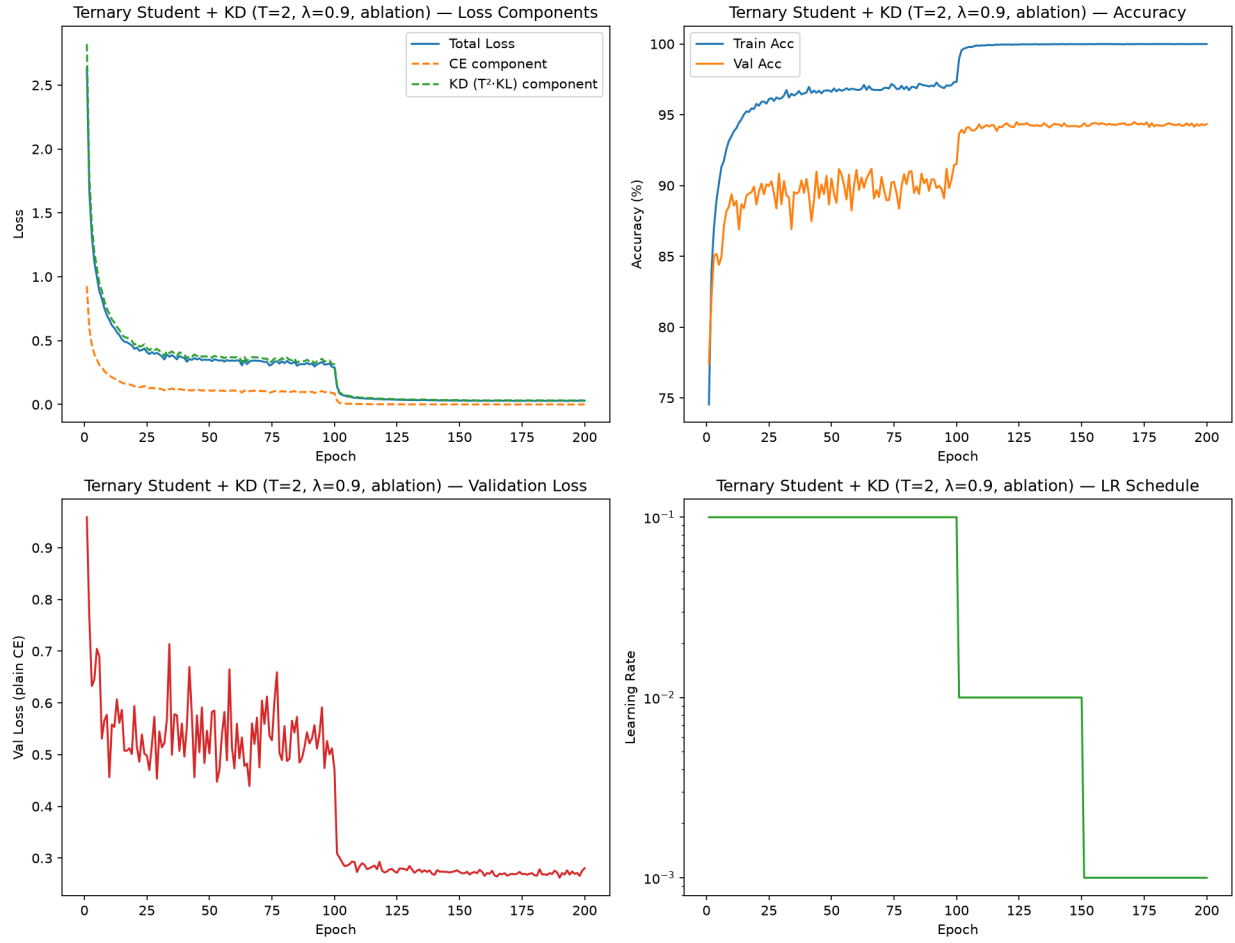


Figure 11: student ablation curves

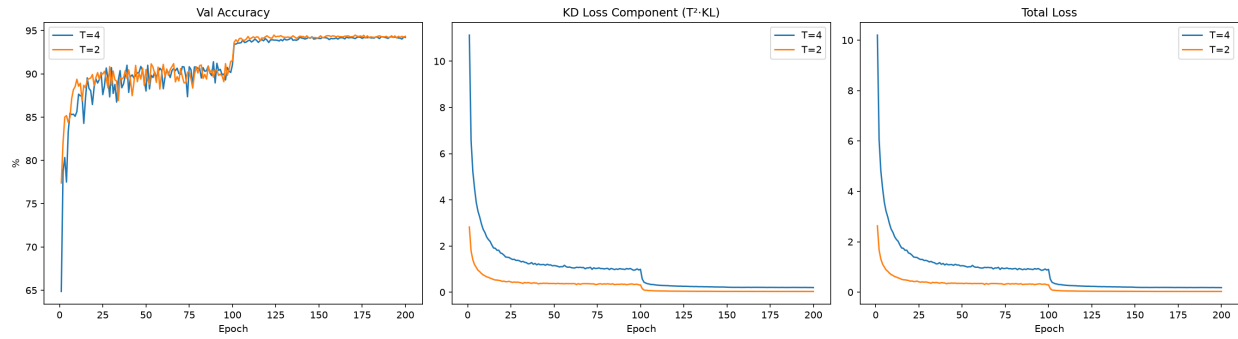


Figure 12: Val Accuracy / KD Loss Component / Total Loss, T=4 vs T=2

5.3 Interpretation

$$\Delta\text{accuracy} = 0.01 \text{ percentage points} \tag{4}$$

The two tested temperatures produce essentially identical final test accuracy, despite the KD loss component and total loss trajectories differing visibly throughout training ($T = 4$ maintains a higher KD loss magnitude throughout, consistent with the T^2 scaling term). Within the tested range, temperature has limited observable impact on final classification accuracy. This does **not** establish an optimal temperature, since only two values were evaluated — a wider sweep would be needed to characterize any non-monotonic trend.

6 Critical Discussion and Limitations

6.1 Accuracy–Compression Trade-off

The ternary student achieves $\approx 15.8\times$ theoretical compression over the FP32 ResNet18 baseline while retaining $\approx 94.3\%$ test accuracy — within 0.1pp of the full-precision baseline and the much larger teacher. This trade-off is favorable, but the compression is achieved under a highly constrained weight representation whose behavior may not generalize identically to other datasets, architectures, or a wider bit-width range.

6.2 Effect of Ternarization Across Layers

Sparsity and α both vary substantially by layer depth (Section 4.4), indicating that ternarization does not affect all layers uniformly. This experiment ternarized **all** convolutional and fully-connected layers, including the first and last. No dedicated layer-sensitivity ablation (e.g. selectively keeping the first/last layer FP32 and measuring the accuracy delta) was performed, so **which specific layers are most sensitive to ternarization cannot be claimed from these results** — only that layer-wise statistics (sparsity, α) differ substantially.

6.3 Role of Knowledge Distillation

KD is intended to transfer the teacher’s softened output distribution to improve the student’s training signal beyond hard labels alone. However, **a ternary student trained without KD was not evaluated** in this study, so the isolated contribution of KD to the ternary student’s final accuracy cannot be quantified — the 94.29% result reflects the combined effect of KD and QAT together, not KD’s effect in isolation.

6.4 Temperature Sensitivity

$T=2$ and $T=4$ produced nearly identical test accuracy (94.28% vs. 94.29%). The tested temperature range had little measurable effect on final classification accuracy in this setup, though this should not be generalized beyond the two values tested.

6.5 Practical Hardware Limitation

The theoretical 2-bit ternary weight representation does not by itself imply a $15.8\times$ (or any) inference speedup. Realizing actual acceleration depends on hardware support for low-bit arithmetic, weight packing, specialized ternary/low-bit inference kernels, memory access patterns, and

implementation overhead — none of which are implemented or benchmarked here. This result demonstrates **storage/representation efficiency**, not measured hardware acceleration.

6.6 Experimental Scope

This study is limited to CIFAR-10 and the ResNet18/34 architecture pair, a single temperature ablation (two values), no ternary-without-KD control, no hardware latency benchmark, and no systematic layer-sensitivity study. The out-of-distribution check on CIFARNet (Section 4.6) showed a drop to 78.0% overall accuracy versus 94.29% in-distribution, indicating the model’s compressed representation generalizes less robustly outside its training distribution — a further scope limitation.

6.7 Scope of Claims Supported by These Experiments

Given the scope of experiments actually run, the following should not be implied elsewhere in this report:

- **No optimal KD temperature is established.** Only $T = 2$ and $T = 4$ were tested; the near-identical result (0.01pp difference) describes this pair only, not the full temperature space.
- **Which specific layer(s) are most sensitive to ternarization cannot be determined.** No ablation selectively exempted individual layers (e.g. first conv, final FC) from ternarization; only aggregate per-layer sparsity/ α statistics were measured (Section 4.4), which describe what happened, not how much it mattered to accuracy.
- **KD’s isolated contribution to the student’s accuracy cannot be quantified.** No ternary student was trained without KD, so the 94.29% result cannot be decomposed into “how much came from QAT” vs. “how much came from KD.”
- **No real hardware inference speedup is demonstrated.** The $15.8\times/30.08\times$ figures (Section 4.5) are theoretical storage calculations only; no low-bit inference kernel was implemented or benchmarked, and actual latency depends on hardware support this study does not test.

7 Conclusion

This work asked: **can ternary QAT combined with knowledge distillation produce a substantially compressed ResNet18 while retaining competitive CIFAR-10 accuracy?** The results support this: the ternary student reaches 94.29% test accuracy, within 0.10pp of the FP32 baseline (94.13%) and 0.10pp of the ResNet34 teacher (94.39%), at a global ternary sparsity of $\approx 59.2\%$ and a theoretical compression of $\approx 15.8\times$ over the FP32 baseline ($\approx 30.08\times$ over the teacher). The KD temperature ablation ($T = 2$ vs. $T = 4$) showed negligible difference (0.01pp) within the tested range. The principal limitation is that this compression is theoretical: realizing genuine inference speedup requires specialized low-bit hardware kernels not implemented or benchmarked in this study.

References

- [1] He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition*. CVPR.

- [2] Hinton, G., Vinyals, O., & Dean, J. (2015). *Distilling the Knowledge in a Neural Network*. NeurIPS Deep Learning Workshop.
- [3] Li, F., & Liu, B. (2016). *Ternary Weight Networks*. arXiv:1605.04711.
- [4] Zhu, C., Han, S., Mao, H., & Dally, W. J. (2017). *Trained Ternary Quantization*. ICLR. (Cited for the restated TWN equations (Eqs. 4–5) used to formulate Section 2.3; TWN itself, not TTQ’s learned-scale variant, was implemented.)
- [5] Bengio, Y., Léonard, N., & Courville, A. (2013). *Estimating or Propagating Gradients Through Stochastic Neurons for Conditional Computation*. arXiv:1308.3432.
- [6] Yin, P., Lyu, J., Zhang, S., Osher, S., Qi, Y., & Xin, J. (2019). *Understanding Straight-Through Estimator in Training Activation Quantized Neural Nets*. ICLR.
- [7] Hubara, I., Courbariaux, M., Soudry, D., El-Yaniv, R., & Bengio, Y. (2016). *Binarized Neural Networks*. arXiv:1602.02830. (Source of the saturated/clipped STE convention actually implemented.)